

The forward-expansion programme: a synthesis (germbij Theorem 3.1’s forward direction, twelve stages)

Tide loop (Claude Fable 5, with GPT-5.6 Sol deliberation)

2026-08-10

Abstract

Twelve stacked tides (PRs #96–#110) carried the germbij note’s Theorem 3.1 from its inverse half — superpolynomially-close localized moment families determine the location, the positive-order jet, and for analytic losses the germ — to the full nondegenerate statement: on a **ForwardExpansionDomain** (a nondegenerate minimum with C^{N+2} regularity and a Peano remainder), the rescaled posterior moment of every continuous observable of polynomial growth *is* an order- N asymptotic polynomial at 0^+ (**rescaledMoment_hasExpansion**), with monomial instances matching the data consumed by the inverse half and a jet-comparison corollary. Expansion data and Taylor data now determine each other in both directions, in Lean. The programme followed a single up-front design consult almost to the letter; its one architectural deviation — replacing the prescribed log-derivative coefficient recursion by $\exp$ ’s Taylor truncation composed with the exponent polynomial — removed every coefficient identity from the analytic spine, and its one analytic correction — an unrestricted exponential remainder in place of Mathlib’s $|x| \leq 1$ bound, with the q -power recovered multinomially — was forced by the geometry of the mesoscopic window, where corrections are small only relative to the quadratic.

1 What was proven

- **AsymptoticPolynomial** (#96): the expansion predicate **IsAsymptoticExpansionTo** at 0^+ and coefficient uniqueness — the tool that later closed both the quadratic bridge and the jet-comparison corollary.
- **GaussianMeso** (#97): the mesoscopic window $\sqrt{q} \|z\| \leq 1$ and its tail calculus — outside the window, polynomial-Gaussian and rescaled-Boltzmann integrals are beyond all orders.
- **ForwardDomain** (#98): the one-field Peano mixin over the merged higher-order package, and the exact exponent split $(L(qz) - L(0))/q^2 = T_2(z) + \sum_{s=1}^N q^s V_s(z) + q^N \rho_q(z)$ with pointwise vanishing and window bounds on ρ .
- **ExpGraded** (#99): the graded expansion of $\exp(-(\sum_s q^s a_s + q^N \rho_q))$ with coefficients from the Taylor-truncation polynomial — three elementary steps, no coefficient identities.

- **CoeffFn** (#100): continuity and polynomial growth of the coefficient functions $z \mapsto P_j(z)$ at arbitrary degree, by induction through the coefficient convolution.
- **ScalarBounds** (#101): the unrestricted exponential Taylor remainder $|e^x - S_N(x)| \leq |x|^{N+1}e^{|x|}/(N+1)!$, the perturbation bound, and the graded polynomial tail with a z -uniform indexing bound.
- **GaussAbsorb** (#102): $T_1 = 0$ and $T_2 = \frac{1}{2}$ qform H as *theorems* (ray coefficient uniqueness), ε -Peano window control, and the absorption of all corrections into $e^{-(\lambda/4)\|z\|^2}$.
- **WindowMajorant** (#103): the single q -uniform majorant the dominated-convergence step consumes.
- **NumeratorExpansion** (#104) and **NumeratorTails** (#106): the window convergence by filter-indexed dominated convergence with the indicator folded in; the two outer tails; and `numerator_hasExpansion` with coefficients $\int P e^{-T_2} P_j$.
- **AsymptoticDivision** (#108): quotients of order- N expansions with nonzero constant denominator coefficient.
- **ForwardTheorems** (#110): partition positivity, the moment expansion, the monomial instances, and jet comparison.

2 Design decisions that carried the programme

Existence-plus-comparison as the public API. The programme never needed explicit coefficient values: the note’s theorem asserts that expansions exist and determine the jet, and stage A1’s uniqueness converts any two expansion claims into coefficient equality. Every collection mechanism stayed internal.

The Taylor-truncation collection mechanism. The design consult prescribed the log-derivative recursion $P_j = -\frac{1}{j} \sum s V_s P_{j-s}$ but noted the proof may use `exp`’s ordinary Taylor remainder. Defining the coefficients as the coefficients of $\sum_{i \leq N} (-A)^i / i!$ made the scalar expansion theorem three elementary $O(q^{N+1})$ steps and moved all combinatorics into a measurability-and-growth induction through `Polynomial.coeff_mul` — the same mechanism then reused for the division stage’s Cauchy product.

The unrestricted remainder over a second window. On the mesoscopic window the exponent correction reaches $\sim 1/\sqrt{q}$; Mathlib’s `Real.exp_bound` ($|x| \leq 1$) cannot see it. The stage-5 consult confirmed the fix: prove $|e^x - S_N(x)| \leq |x|^{N+1}e^{|x|}/(N+1)!$ once, and recover the q -power from $|A| \leq q \sum_s |V_s|$ on $q \leq 1$ rather than from absolute smallness. A smaller window would have needed its own tail theorems and still failed at $N = 0$.

The quadratic bridge by self-consumption. The domain ties H to L only through the quadratic Peano condition, so $T_2 = \frac{1}{2}$ qform H was proven by composing both Peano statements along rays and applying the programme’s own stage-1 uniqueness — which also yielded the critical-point condition $T_1 = 0$, converting the exponent split’s hypothesis into a theorem.

One majorant, one DCT. All domain-specific uniformity concentrated in a single theorem (`normalized_window_remainder_bound`); the dominated convergence then ran with the window indicator folded into the integrand, per the consult, and the final packaging was four subidentities closed by linear arithmetic.

3 Process notes

Two architecture consults (the seven-stage programme design; the stage-5 shape with its nine-step implementation order) were followed essentially verbatim, and both are archived in the tide log. Every stage’s interface was consumed by the next without rework; the majorant — the longest single proof — compiled on its second build attempt. Recurring Lean friction was almost entirely catalogued classes: beta-redex goals from `integral_congr_ae` (now: `beta_reduce` first), norms on both sides of `isLittle0_iff`, `positivity` blind to opaque structure fields, and two deterministic whnf heartbeat timeouts. New catalogue entries from this arc are recorded in the per-tide retrospectives.

4 What remains

Of the germbij note: the semi-quasi-homogeneous constructive class (the M_α recovery, Gelfand–Leray, and the weighted triangular induction), ranked less crucial in the commissioning direction; and, if ever wanted, the connection of the abstract `momentCoeff` to explicit Wick/Isserlis values through the Gaussian moment integrals already in the seabed. A parallel fidelity review (PR #105) flags the inverse-half headline as conditional on a matched 0–2 jet; its perimeter debt list is the right starting point for any tightening pass.